

Name: Sagi Sahasra Srihasa
Roll : 2520030008

Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_COMMAND 100

// Node for command history
typedef struct Node {
    char *command;
    struct Node *next;
    struct Node *prev;
} Node;

Node *head = NULL;
Node *tail = NULL;

// Add command to history
void add_history(const char *cmd) {
    Node *newNode = malloc(sizeof(Node));

    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return;
    }

    newNode->command = malloc(strlen(cmd) + 1);

    if (newNode->command == NULL) {
        free(newNode);
        printf("Memory allocation failed\n");
        return;
    }

    strcpy(newNode->command, cmd);

    newNode->next = NULL;
    newNode->prev = tail;
```

```

    if (tail != NULL)
        tail->next = newNode;
    else
        head = newNode;

    tail = newNode;
}

// Display history
void display_history() {
    Node *temp = head;
    int count = 1;

    while (temp != NULL) {
        printf("%d %s\n", count++, temp->command);
        temp = temp->next;
    }
}

// Free all memory
void free_history() {
    Node *temp = head;

    while (temp != NULL) {
        Node *next = temp->next;

        free(temp->command);
        free(temp);

        temp = next;
    }

    head = NULL;
    tail = NULL;
}

```

```

int main() {
    char input[MAX_COMMAND];

    printf("Mini Shell History\n");
    printf("Type 'history' to display commands.\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("shell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        // Remove newline
        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strcmp(input, "history") == 0) {
            display_history();
            continue;
        }

        if (strlen(input) > 0) {
            add_history(input);
        }
    }

    free_history();

    return 0;
}

```

Output:

```
shell> exit
sahasrasagi@Sahasra:~/OS_Lab/skill$ nano skill_3.c
sahasrasagi@Sahasra:~/OS_Lab/skill$ ./skill_3
Mini Shell History
Type 'history' to display commands.
Type 'exit' to quit.

shell> history
shell> exit
sahasrasagi@Sahasra:~/OS_Lab/skill$ |
```